INCH-POUND
MIL-R-22388A(OS)
20 June 1989
SUPERSEDING
MIL-R-22388(Wep)
10 March 1964

MILITARY SPECIFICATION

RESOLVER, BALL, 1 INCH, ASSEMBLY

This specification *is approved for use by the Naval Sea Systems Command, Department of fhe Navy and is available for use by all Departments and Agencies of the Department of Defense*

1. SCOPE

1.1 Scope *This* specification describes the requirements for the procurement of one type of Ball Resolver BUWEPS LD-498192. The ball resolver is a precision mechanical computing device having two angular Inputs and two angular outputs with the angular outputs equal to 1.65 times the integral of the sine or cosine of one angular input with respect to the angular input.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards The following specifications and standards forma part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

UU-P-268	Paper, Kraft, Wrapping
PPP-B-601	. Boxes, wood, cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-corner
PPP-B-636	Boxes, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation, Method of
MIL-S-401	Shock Tests, H.I (High-impact); Shipboard Machinery. Equipment andi Systems, Requirements for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardisation Branch (3730), Indian Head, Maryland 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC No. N/A FSC 1290

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MIL-L-10547	Liners, Case, and Sheet, Overwrap: Water-Vaporproof or waterproof, Flexible
MIL-F-18870	Fire Control Equipment, Naval Shipboard, General Specification for

STANDARDS

FEDERAL

FED-STD-H28	Screw-thread standards for Federal services
FED-mI-Iot	Test Procedures for Packaging Materials

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-167-1	Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited)
MIL-STD-81O	Environmental Test Methods and Engineering Guidelines

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Naval Publications and Forms Center, (Attn: NPODS), 5801 Tabor Avenue Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents drawings and publications. The following other Government documents drawings, and publications forma part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the soliciation (see 6.2).

DRAWINGS

NAVAL SEA SYSTEMS COMMAND (CAGE Code 10001)

LD48192	Resolver Assembly Ball 1“
SA-2530406	Ball Resolver Test Fixture
SA-2530491	Aligning Pin

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND

OS 6341	Miscellaneous General Ordnance Design Requirements
0012353	Description and Operating Instructions for Ball Resolver Test Fixtures Shop Accessory SA 2530406

(Application for copies should be addressed to Commanding Officer, Naval Ordnance Station, (Attn: Code 802), Louisville, Kentucky 402144001.)

2.2 Non-Government publications. The following documents forma part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed *in the issue* of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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UNIFORM CLASSIFICATION OOMMIITEE PUBLICATION

Consolidated Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago 6, IL)

AMERICAN TRUCKING ASSOCIATION, INC, PUBLICATION

Motor Freight Classification Rules

(Application for copies should be addressed to the issuing Officer, American Trucking Association, Inc., 1424 16th Street, N.W., Washington 6, DC)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified in the contract or purchase order (see 6.2), first article samples of the ball resolver shall be manufactured using the methods and procedures proposed for the production lot. These samples shall be tested as specified in 45 herein and are for the purpose of determining that the production item meets the requirements of this specification and the design.

3.2 Materials. Unless otherwise specified in the contract, requisition, or order (see 6.2), all materials and purchased parts entering into the final assembly of the ball resolver assembly shall conform to the detailed require ments given on the drawings and specifications listed in BUWEPS LD-498192 and in MIL-F-18870.

3.3 Design and construction.

3.3.1 Finishes, coatings, construction, and dimensions Finishes, coatings, construction, and dimensions for the ball resolver shall be in accordance with the applicable drawings, specifications, and publications listed in BUWEPS LD-498192.

3.3.2 Interchangeability. Unless otherwise specified on the applicable drawings, component parts of the assembled units fabricated in accordance with their applicable drawings, shall be interchangeable and shall assemble with mating components of the units without selection or fitting. If, during the assembly of the fabricated parts of a unit, selection or fitting is required, then changes in dimensions are necessary, and revision and correction of the applicable drawings are required.

3.3.3 Threads Unless otherwise specified (see 6.2), ail threads shall be in accordance; with FED-STD-H28. The class or fit for threads shall be as specified on the applicable drawings and the NUT-GO gaging practice shall be as set forth in FED-STD-H28.

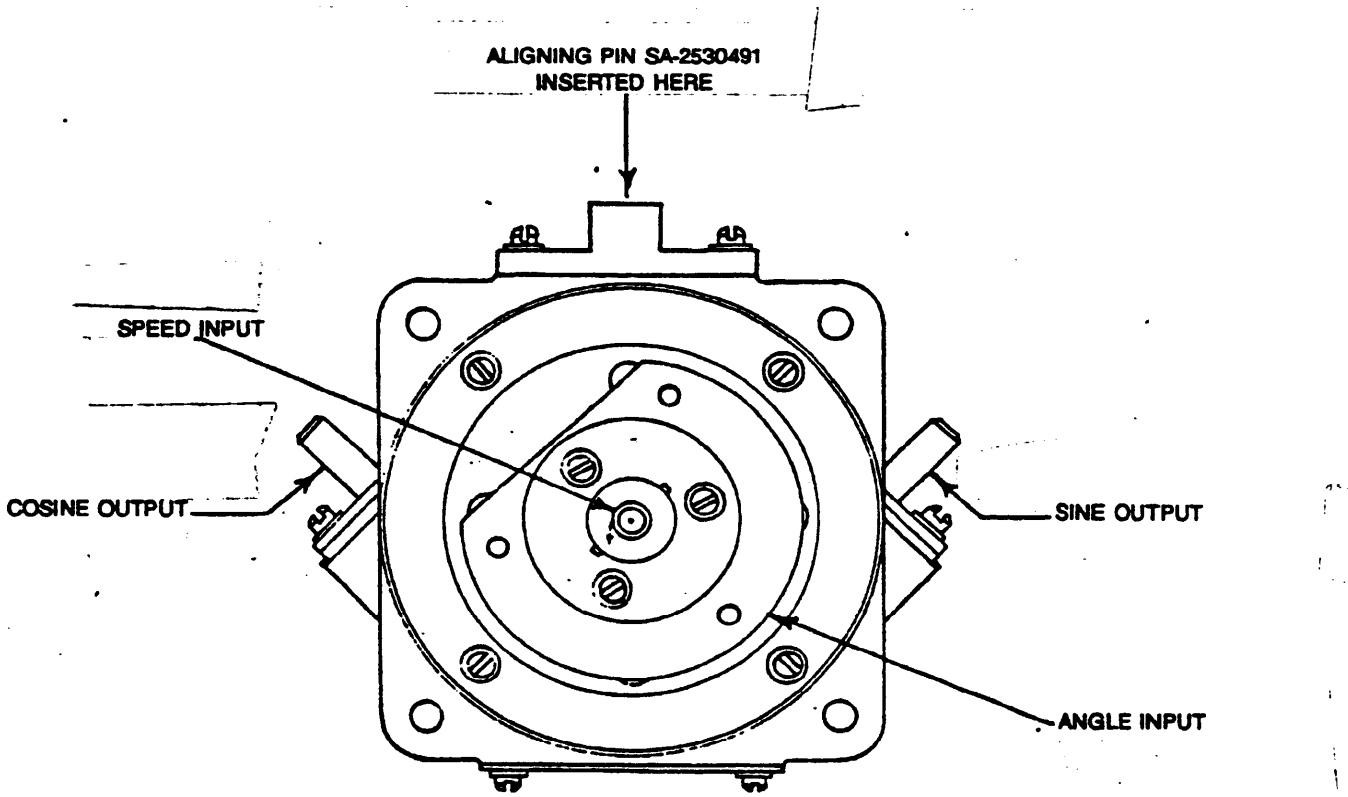
3.4 Moisture fungus and corrosion. The ball resolver shall be rendered moisture fungus, and corrosion resistant in accordance with detail requirements specified on drawings and specifications in LD-498192.

3.5 Marking. Markings of assemblies, subassemblies, and parts shall be in accordance with the applicable drawings listed in LD-498192.

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3.6 Performance requirements and product characteristics. The ball resolver assembly shall meet the following performance requirements and product characteristics:

3.6.1 Alignment When tested in accordance with 4.6.1.2, the zero position of the angle input shaft (see figure 1) shall be such that when the speed input shaft is rotated at the rate of 300 revolutions per minute (rpm) for 1000 plus or minus 0.25 revolutions,the sine output shaft shall rotate O plus or minus 2.0 revolutions and the cosine output shaft shall rotate 1650 plus or minus 2.0 revolutions The zero position of the angle input shaft of the ball resolver is that angular position which will allow insertion of Aligning Pin SA-2530491.



(Angular input shaft shown in zero position)

FIGURE 1. *Resolver, ball, 1 inch, assemby*

3.6.2 Accuracy. When tested in accordance with 4.6.1.2 and for the 12 specific test conditions listed table 1, the accuracy of the ball resolver shall be such that none of the 24 errors in output shaft angle exceeds 5.0 rpm and the arithmetic of the 24 output errors does not exceed 2.0 rpm.

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Table I. *Alignment test conditions.*

Angle input (degrees) tolerance plus or minus 30 seconds	Speed input (rpm)[1] tolerance plus or minus 0.25 rpm	Sine output (rpm)	Cosine output (rpm)
15	1000	427'	1593.7
45	1000	I166.7	1166.7
75	1000	1593.7	427.0
105	1000	15937	427.0
135	1000	1166.7	1166.7
165	1000	427.0	1593.7
	1000	427.0	1593.7
225	1000	1166.7	11607
255	1000	1593.7	427.0
285	1000	1593.7	427.0
315	1000	1166.7	1166.7
345	low	427.0	1593.7

[1]At a rate of approximately 300 revolutions per minute.

3.7 Environmental requirements. The ball resolver shall meet the applicable requirements of this specification during or after tests defined or specified in 4.7.

3.7.1 Vibration, nonoperating condition. The ball resolver shall withstand the vibration tests specified in 4.7.1 and thereafter shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

3.7.2 Shock, nonoperating condition. The ball resolver shall withstand the shock tests specified in 4.7.2 and thereaftershall show no damage and shall meet all the requirements and product characteristics of this specification.

3.7.3 Temperature, operating condition. The ball resolver shall be subjected to temperature tests of 4.7.3 and thereafter shall meet all the performance requirements and product characteristics of this specification.

3.7.4 Temperature, nonoperating condition. The ball resolver shall be subjected to temperature tests of 4.7.4 and thereafter shall meet all the performance requirements and product characteristics of this specification.

3.7.5 Humidity. The ball resolver shall be subjected to humidity tests of 4.75 and thereafter shall meet all the performance requirements and product characteristic of this specification.

3.7.6 Tropical environment. The ball resolver shall be exposed to fungus tests of 4.7.6 and thereafter shall meet ail the performance requirements and product characteristics of this specification.

3.7.7 Vibration - transportation. The ball resolver, after being subjected to the tests specified In 4.7.7, shall show no damage and shall meet all the requirements of section 3.

3.7.8 Shock transportation, storage, and handling. The ball resolver, after being subjected to the tests specified in 4.7.8 shall show no damage and shall meet all the requirements of section 3

3.8 clean-up. Prior to and after final assembly, all parts, components, and the assembly shall be thoroughly cleaned of loose metal chips, and other foreign matter. Burrs and sharp edges shall be removed.

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3.9 Workmanship. The equipment, including all parts and accessories, shall be constructed and finished in a manner to assure compliance with all requirements of this spcification. Particular attention shall be paid to marking of parts and assemblies, machine screw assemblage, and freedom of parts from burrs and sharp edges.

3.9.1 First ● rticle workmanship. The workmanship exhibited in the first article samples will be evaluated to determine acceptability. The approved standards of workmanship will thereby become a minimum production requirement for ail units offered for acceptance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contractor purchase order (see6.2), the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order (see 6.2), the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5 The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Test and inspecction equipment and facillities The manufacturer shall furnish and maintain all necessary test equipment, facilities, and personnel for performing all acceptance tests The test equipment shall be adequate in quantity, and unless definite requirements are specified, shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

4.1.2.1 Gages. The contractor shall provide himself with whatever gages are necessary and adequate to insure that the material to which this specification applies will meet the requirements shown on applicable drawings. if so stipulated in the contract, order, or requisition (see 6.2), the Government will furnish drawings of pertinent Navy Final inspection Gages for guidance in design of the contractor's inspection gages. However, such procedures shall not relieve the contractor of his responsibility in the design and manufacture of such gages as may be required for the satisfactory fulfilment of the contract requirements but is intended to facilitate acceptance of all components and assemblies by Navy Find Inspection Gages.

The following items of test equipment are required to perform the acceptance tests set forth in this specification:

a Shop Accesssary SA-2530406 Ball Resolver Test Fixture
b. Shop Accessory SA-2530491, Aligning Pin

4,2 Test after delivery. The government reserves the right to conduct any or all tests specified herein as deemed necessary after installation of the ball resolver. Any defects discovered during or after these tests due to the contractor's failure to comply with drawings or specifications for this equipment shall be corrected by the contractor at no expense to the government, provided the contractor is notified of such defects within the guarantee period of the contract.

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4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 45).
b. Quality conformance inspection (see 4.6).

4.4 Test conditions. unless otherwise specified, the ball resolver assembly shall be subjected to acceptance tests under the following conditions:

a. Temperature - Room ambient 65 to 95 degrees Fahrenheit °F) (18 to 35 degrees Celsius (°C)
b. Altitude — Normal ground atmospheric pressure
c. Vibration — None
d. Humidity — Room ambient to 95 parent relative maximum.

4.5 Flint article inspection. Unless otherwise specified for each contractor purchase order, a representative sample of three ball resolvers shall be manufactured by the methods and procedures proposed for the production lot. These shall be submitted for first article tests at an activity designated in the contractor by the Naval Sea Systems Command, for purposes of determining compliance with the requirements of the contract, specification, and drawing Further production of the item by the contractor prior to approval of the contracting activity or the completion of tests on the first article samples shall be at the contractor's risk. First article samples accepted will be applied as part of the quantity specified by the contract or order but shall not be used in any equipment manufactured for the Government.

4.5.1 Environmental testing. Environmental tests shall be accomplished on samples selected as specified in 4.5, which are representative of the production of the ball resolver after the award of the contract, to determine that the production meets the requirements of this specification. These tests are detailed in 4.7. Acceptance shall be based on m defects in the sample Failure of the sample to comply with these requirements will result in the cessation of production, as determined by the contracting activity.

4.6 Quality conformance inspection. Upon approval of the first article, production units shall be sampled and tested according to 4.6.1 and 4.6.2 for conformance to this specification.

4.6.1 Performance and characteristics inspection. Each ball resolver shall be tested as specified in 4.6.1.1 and 4.6.1.2

4.6.1.1 Visual inspection. Each ball resolver shall be examined and measured to assure compliance with 3.2, 3.4, 3.5,3.8,3.9, and section 5. Any defects shall be cause for rejection of the unit.

4.6.1.2 Performance test. The ball resolver shall be tested in accordance with the detailed test procedure given in OD 12353 to determine compliance with the requirements 3.3,3.6 and 3.7.

4.6.2 Environmental testing. Environmental tests shall be accomplished on samples selected as specified in 4.6.2.1, which are representative of the production of the ball resolver after the award of the contract, to assure that the production meets the requirements of this specification. These tests are detailed in 4.7. Acceptance shall be based on m defects in the sample Failure of the sample to comply with these requirements will result in the rejection of the lot or the cessation of production, as determined by the contracting activity.

4.6.2.1 Periodic production samples. Unless otherwise specified by the contracting activity, environmental tests samples shall be selected from production as follow

Step 1. Upon initiation of production, select 1 unit from each group of 10 units for submission to an activity designated by the contracting activity.

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Step 2. Upon satisfactory compliance of each unit from *3 consecutive* groups, select 1 unit from each group of 25 units.

Step 3. Upon *satisfactory* compliance of each unit from *3 consecutive* groups, select 1 unit from each group of 50 units. Continue this procedure until a defective unit is found.

Step 4. When a defective unit is found in any group, the group represented by the sample shall be rejected. Shift to step 1 and proceed as indicated.

4.7 Envlronmental tests.

4.7.1 Vbration, nonoperating condition. The ball resolver, rigidly mounted in a suitable vibration fixture, shall be subjected to the Type I vibration tests of MIL-STD-167-1 and shall meet all the performance requirements and product characteristics of this specification after the vibration tests (see 3.7.1).

4.7.2 Shook, nonoperating condition. The ball resolver, rigidly mounted in a suitable shock fixture, shall be subjected to the Type A shock tests In MIL-S-901, and thereafter shall meet all the performance requirements and product characteristics of this specification (see 37.2). -

4.7.3 Temperature, operating condition. The ball rsolver shall be tested to the operational ambient temperatures of between O 'F (-18°C) and 185 'F (85 °C) in accordance with MIL-F-16870 shall function during and after temperature tests, and thereafter shall show no damage or corrosion and shall meet ail the performance requirements and product Characteristics of this specifcation (see 3.7.3).

4.7.4 Temperature nonoperating condtion. The ball resolver shall be tested to storage temperatures Of between -40 'F (-40 °C) and 180 'F (71 'C) in accordance with Condition II of OS 6341 and thereafter shall show no Corrosion or damage and shall meet all the performance requirements and product characteristics of this specification (see 3.7.4).

4.796 Humidity. The ball resolver shall be tested to operational ambient humidities in accordance with MIL-F-18870 and thereafter shall show no corrosion or damage and shall meet ail the performance requirements and product characteristics of this specification (see 375).

4.7.6 Tropical environment. The ball resolver shall be tested to the tropical environmental conditions specified in MIL-SID-810, test method 508, and thereafter shall show no fungus growth, corrosion, or other damage and shall meet the performance requirements and product characteristics of this specification (see 3.7.6). Tropical environmental is defined as conditions of high humidity, high temperature and spore exposure which present an opportunity for fungi growth.

4.7.7 Vibration - transportation. The ball resolver, packaged and packed, as specified in section 5, shall be vibration tested as specified in FED-STD-1OI, Method 5020 (see 3.7.7).

4.7.8 Shock - transportation, storage, and handling. The ball resolver packaged and packed as specified in section 5, shall be subjected to the appropriate rough handling tests specified in FED-STD-101, Methods 5005, 5007, 5008, 5012, and 5014 (see 3.7.8).

4.8 Functional tests. At the conclusion of the tests specified in 4.7.7 and 4.7.8, functional and other tests, as applicable, shall be conducted a the ball resolver to determine freedom from operational malfunction and compliance with all the requirements of section 3

4.9 Waiver of tests. Tests specified in 4.7.7 and 4.7.8 may be waived by the contracting activity for level C packs only, provided the contractor guarantees safe delivery to the using activity

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4.10 Failure of the ball resolver to meet any of the requirements and tests of this specification shall be considered cause for rejection.

5. PACKAGING.

5.1 Preservation. Preservation shall be levels A or C as specified (see 6.2.1).

5.1.1 Level A.

5.1.1.1 Unit packaging. Unit packaging of the ball resolver shall be in accodance with Method Ile of MIL-P-116. The unit package shall contain one ball resolver packaged in a fiber box in accordance with PPP-B-636, Class 2 with Kraft paper overwrap in accordance with UU-P-268.

5.1.1.2 Cushioning. The unit package and pack shall incorporate sufficient cushioning material, bracing, or other adequate shock absorbing devices to ensure that the ball resolver will meet all the requirements of section 3 after being tested in accordance with 4.7.7 and 4.7.8. At the time of packaging or packing, the moisture content of the cushioning material shall be no more than 5 percent. The pH of the material shall be between 6.0 and 8.0.

5.1.2 Level C.

5.1.2.1 Unit packaging. Unit packaging of the motor generator shall be in accordance with Method Ill of MIL-P-116 The unit package shall contain one ball resolver.

5.2 Packing. Packing shall be levels A, B or C as specified (see 6.2.1).

5.2.1 Level A.

5.2.1.1 Exterior containers._ Ball resolvers packaged as described in 5.1.1 shall be packed in a wood box in accordance with PPP-B-601, ovenseas type, or PPP-B-621, Class 2 (gross weight not to exceed 150 lb) or in fiber boxes in accordance with PPP-B-636 Class 2 (gross weight not to exceed 70 lb).

52.1.2 Case liners. Wood exterior containers shall be provided with case liners in accordance with MIL-L-10547.

5.2.2 Level B.

5.2.2.1 Exterior containers. Ball resolvers packaged as described in 5.1.1 shall be packed in a wood box in accordance with PPP-B-601, domestic type or PPP-B-621, Class 1 (gross weight not to emceed 150 lb) or in fiber boxes in accordance with PPP-B-636 Class 1 (gross weight not to exceed 70 lb).

5.2.2.2 Case liners. Wood exterior containers shall be provided with case liners in accordance with MIL-L-l0547.

5.2.3 Level C Ball resolvers packaged in accordance with 5.1.2 shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Uniform Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Normal markings. in addition to the markings required by contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129, including serial numbers.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use The ball resolver is intended for use in the Angle Solvers Mark 18 Mods *2 and 3 the Fire Control* System Mark 112 Mod 1, the Sonar Receiving Sets AN/BQG-1, -2, 3,-4, or other suitable application.

6.2 Acquistion requirements Acquisition documents must specify the following:

a. Title number., and date of this specification

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.2)

c. Quantity required (see 45 and 6.7)

d When first article samples are not required (see 3.1, 4.5, and 6.3)

e. Any special materials and parts to be used (see 3.2)

f. If threads not in accordance with FED-STD-H28 are required (see *3.3.3.*

g.. *The* drawings of pertinent Navy Final Inspection Gages to be furnished by the Government (see 4.1.2.1)

h. If acceptance test conditions in 4.4 should be changed

i. Activity to perform environmental tests on periodic production samples (see 4.6.2.1)

j. Level of preservation required (see 5.1 and 6.4)

k. Level of packing required (see 5.2 and 6.4)

l. Any special marking required (see 5.3.1)

m. Whether packaged unit tests can be waived (see 4.7).

6.3 First article When a first article inspection is required, the items should be a first article sample. The first article should consist of three units The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination approval of first article test results, and disposition of first articles invitations forbids should providethat the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Criteria for use of the proper level of preservation, packaging, and packing are as follows:

For Level A — This level is used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhera

For Level B - This level is used only when it is definitely known that the item will be held in covered storage in overseas locations for 6 months or lees or in domestic locations for an indefinite period.

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For Level C — This level is used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

6.5 The attention of the contracting officer is invited to the Quality Assurance Provisions and options in MIL-P-116 and to paragraph 4.9.

6.6 Guarantee. The contractor purchase order should include a requirement for a guarantee stating that any ball resolver which fails to meet the requirements of this specification within 1 year after delivery because of a defect in material or workmanship or because of the vendor's failure to comply with the requirements of the applicable drawings and specifications shall be replaced by the vendor at no cost to the Government.

6.7 The contract or purchase order should include a sufficient number of extra items to cover production and first article testing samples.

6.8 Changes from previous issure. astericks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes

NOTICE When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever, and the fact the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented inventions that may in any way be related thereto.

6.9 Subject term (key word) listing.

Angle solver
Computing device
Fire control system

Preparing activity
Navy-OS

(Project 1290-N428)

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-22388A (OS)	RESOLVER, BALL, 1 INCH, ASSEMBLY

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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